Number patterns

Teacher resource

Question 1

3, 4, 7, 16, ?

The rule for this number pattern can be described as:

next number = *previous number* x 3 -∇, where ∇ represents a certain number.

What is the next number in this pattern?

Answer: 43
Skill: Students recognise and continue a number pattern involving two operations.

Additional questions

1. If the first term in this number pattern was 2, list the first five terms of the pattern.
2. If the first term in this number pattern was 0, list the first five terms of the pattern.
3. The rule for a number pattern is: *next number* = *previous number* x -3 – 5.
   If the first term in this number pattern is 0, write the next three terms.

Question 2

Sam made a pattern of equations.

Number of equation	Equation
1	1 x 2 = 2
2	2 x 3 = 6
3	3 x 4 = 12
4	4 x 5 = 20
5	5 x 6 = 30

What would be the value of equation 20 if this pattern was continued?

Answer: 20 × 21 = 420
Skill: Students recognise, continue and generalise a pattern.

Additional questions

1. Using only one variable, write a rule that determines the value of an equation if you know its equation number.
2. What is the value of equation 1 000 in this pattern?

Question 3

$7a + 1$, $5a + 1$, $3a + 1$, $a + 1$, ______

What would be the next term if this pattern was continued?

O a - 2

O -a + 1

O -a - 1

O -2a + 1

Answer: B
Skill: Students recognise, continue and generalise a pattern.

Additional questions

1. Continue the pattern for three more terms.
2. Evaluate the eight terms of this pattern if $a = 3$. Check if the terms decrease by $2a$.

Question 4

30, 36, 42, 48, 54, ?

In this sequence, 30 is the first term and 36 is the second term.

What is the 201st term in this pattern?

Answer: 1 230
Skill: Students continue and generalise a linear relationship.

Additional questions

1. Describe, in words, the rule to find any term in this sequence.
2. Represent this rule in numbers and symbols.
3. Will the number 5 203 be in this sequence?

Student activity: Number patterns

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where ∇ represents a certain number.

What is the next number in this pattern?

Question 3

7*a* + 1, 5*a* + 1, 3*a* + 1, *a* + 1, _______

What would be the next term if this pattern was continued?

O a - 2

O -a + 1

O -a - 1

O -2a + 1

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